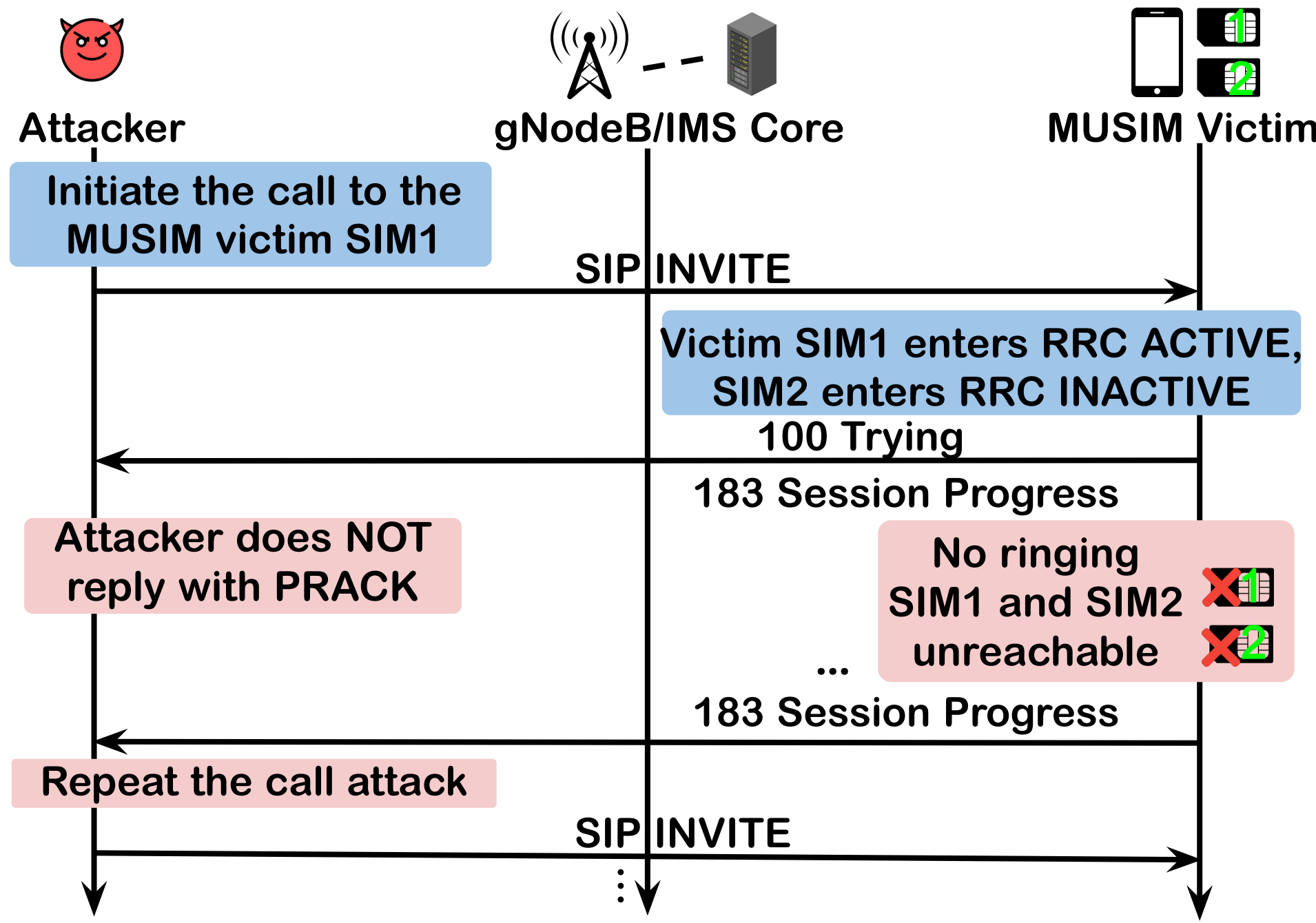
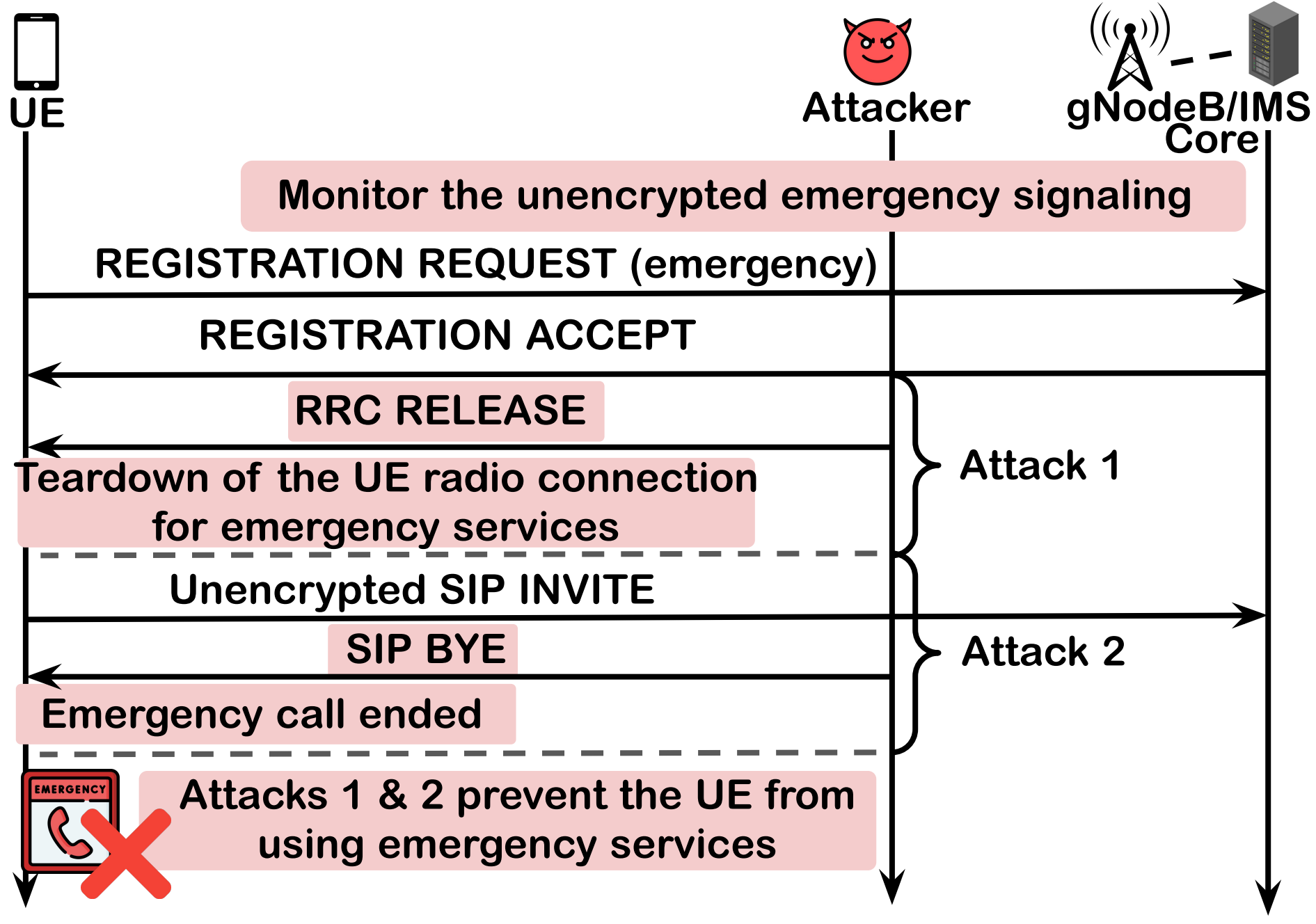
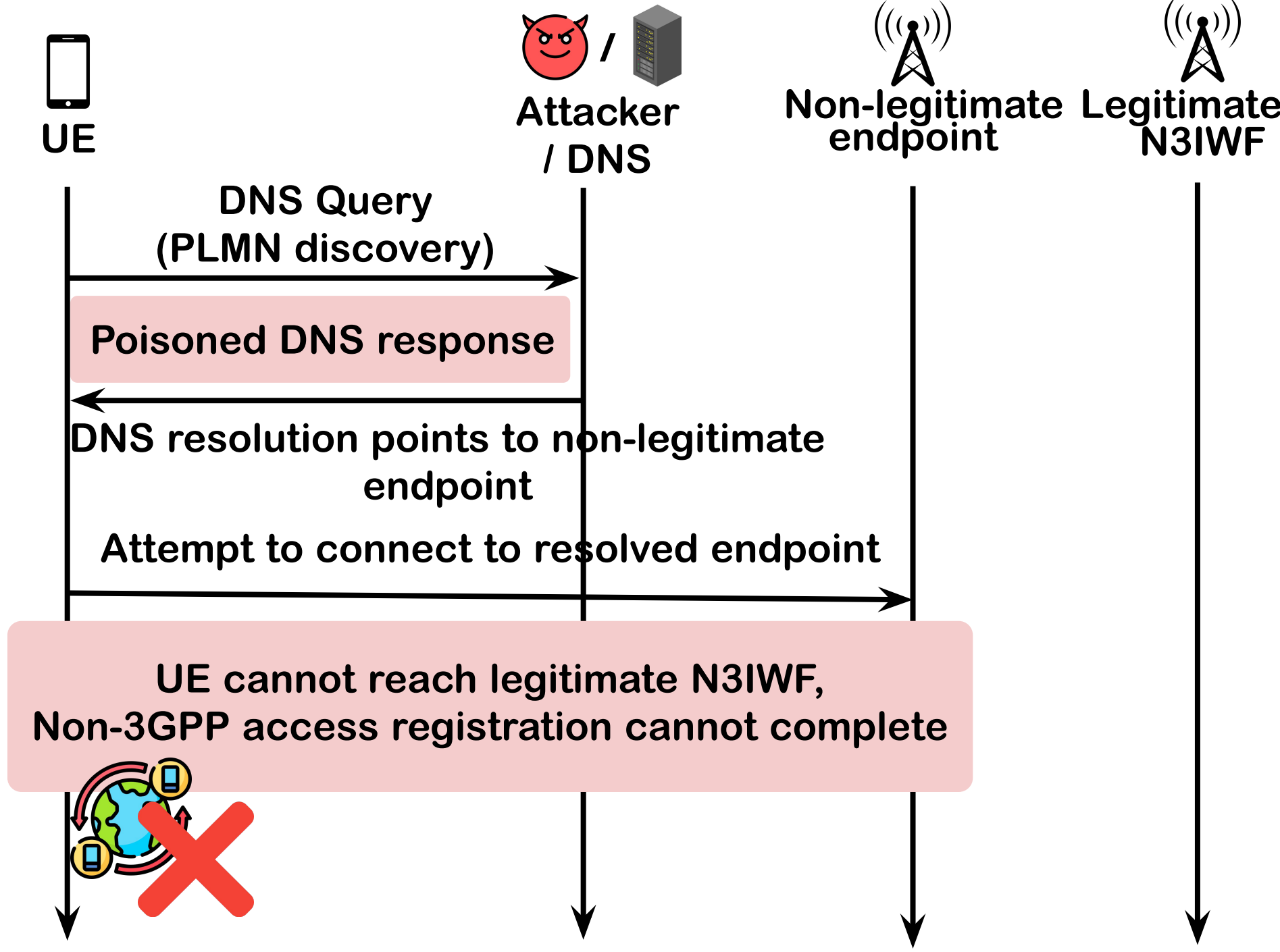
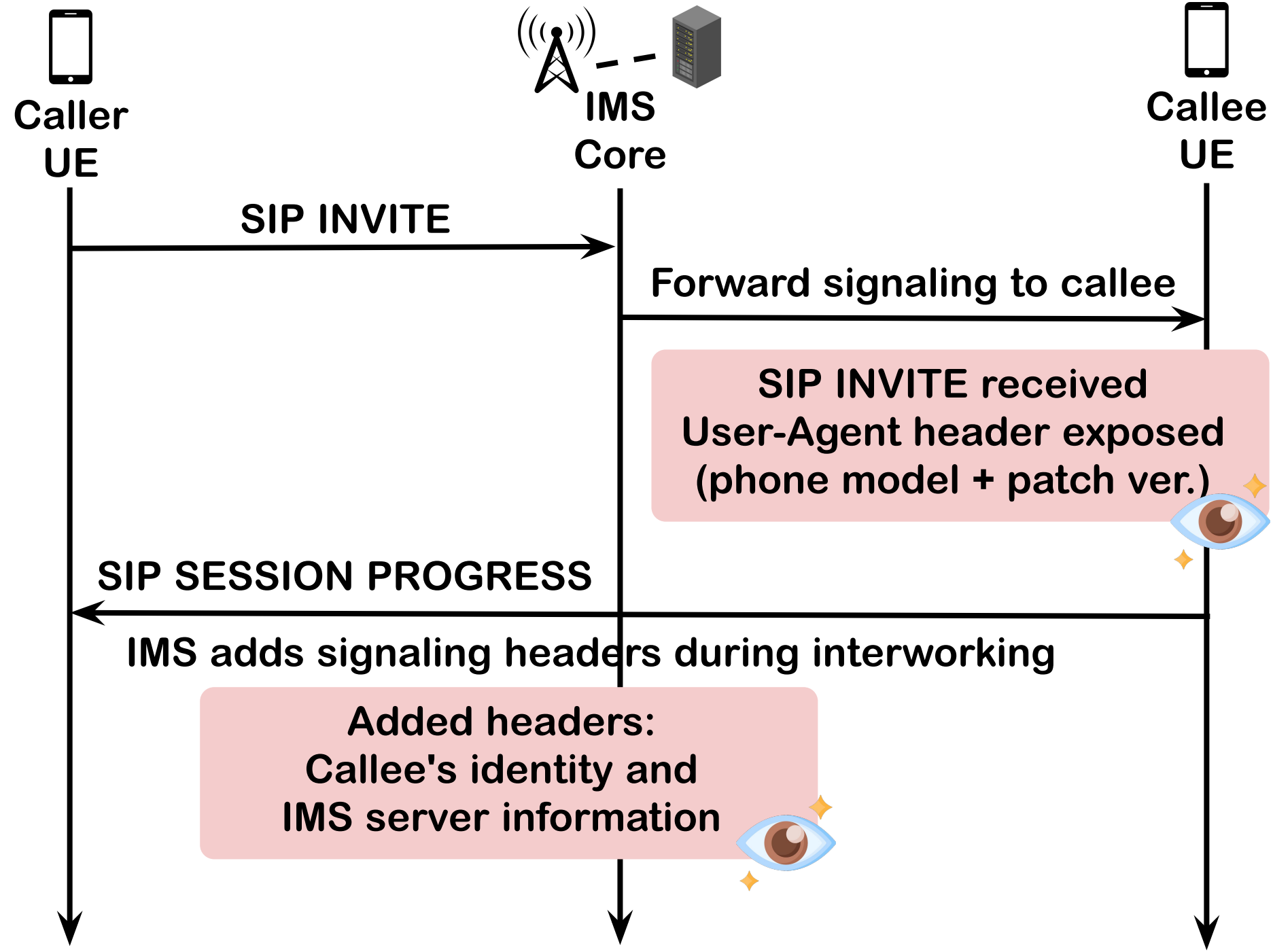
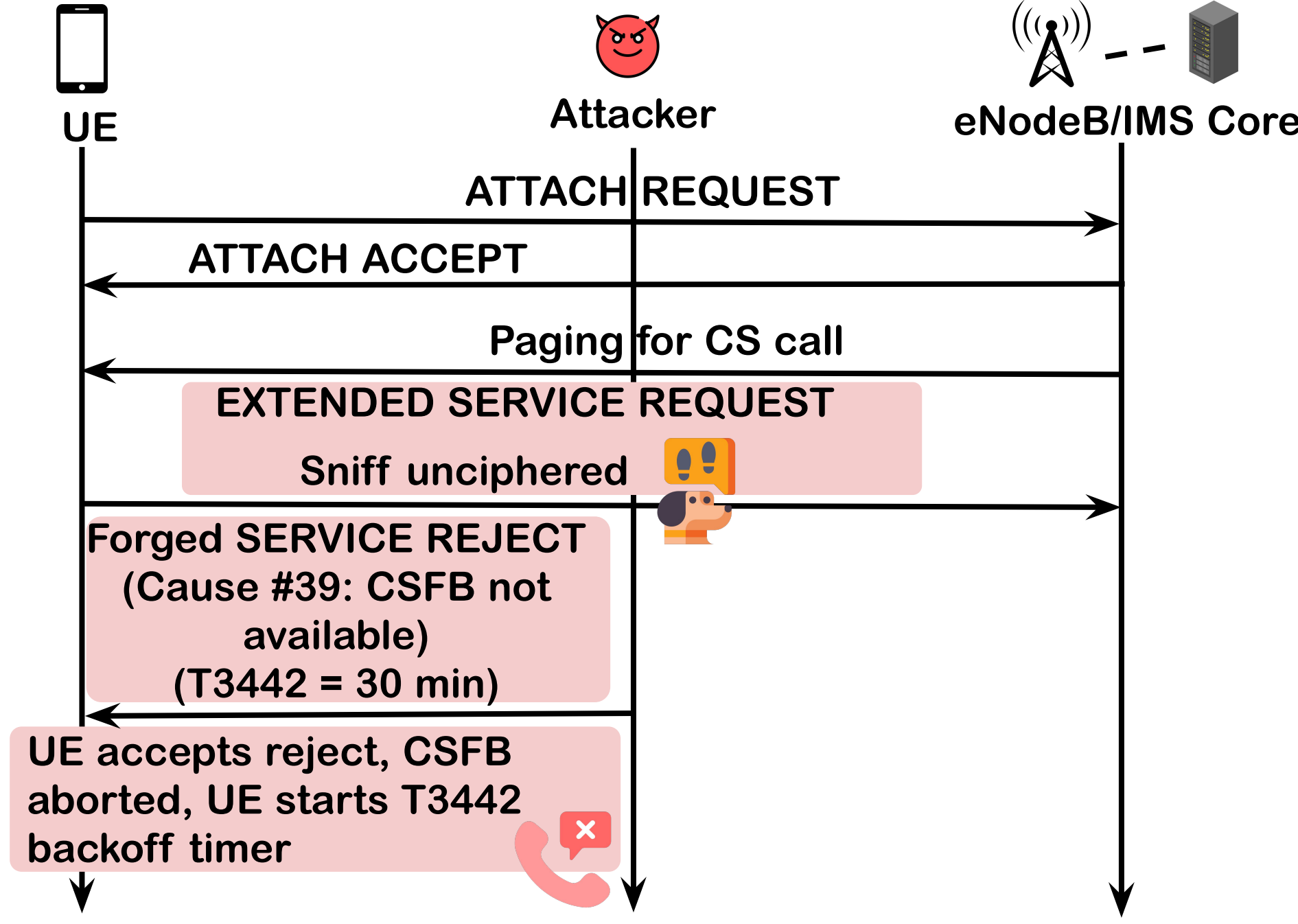
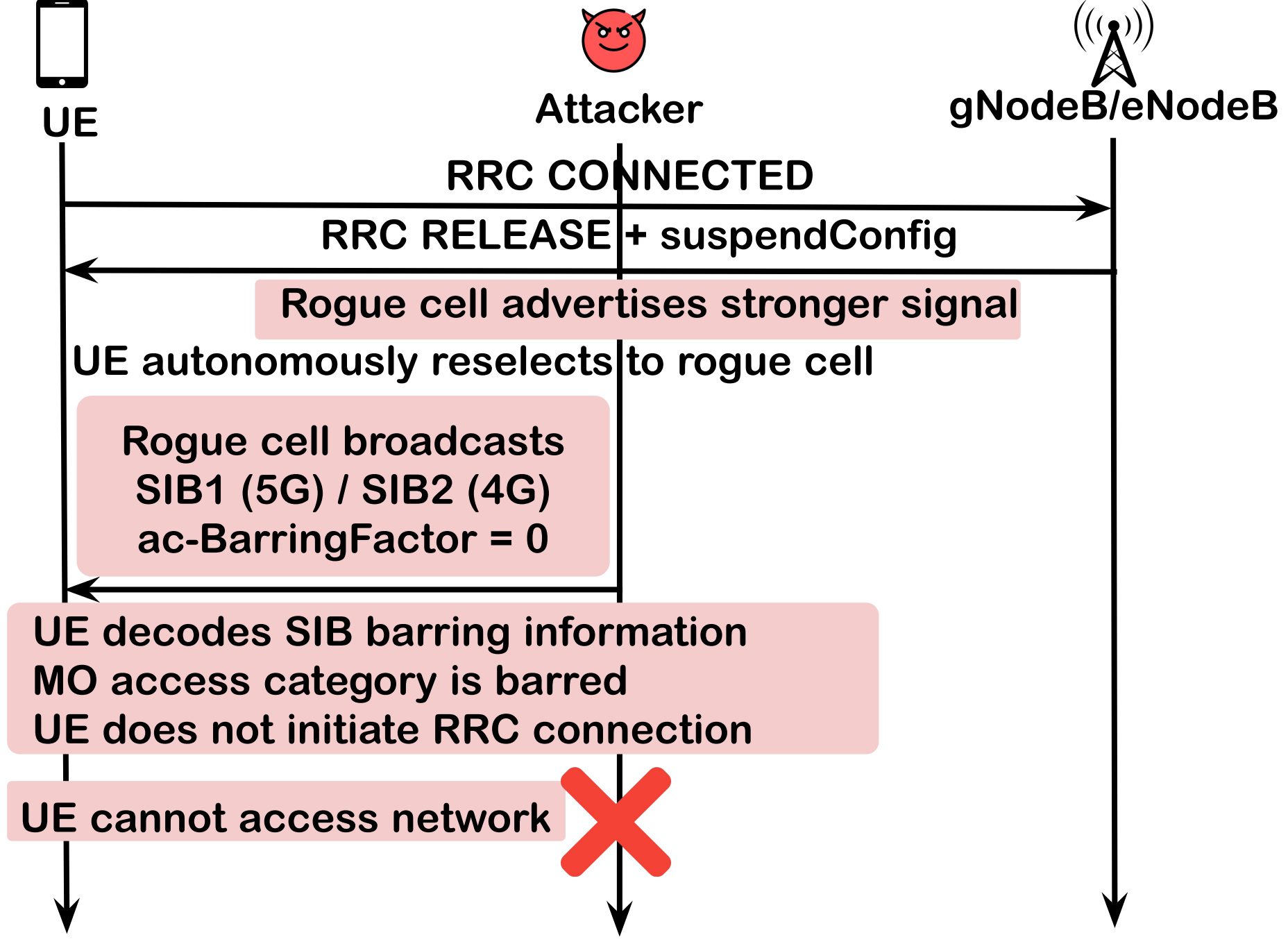

Attacker
gNodeB/IMS Core
MUSIM Victim
Initiate the call to the MUSIM victim SIM1
SIP INVITE
Victim SIM1 enters RRC ACTIVE, SIM2 enters RRC INACTIVE
100 Trying
183 Session Progress
Attacker does NOT reply with PRACK
No ringing SIM1 and SIM2 unreachable
...
183 Session Progress
Repeat the call attack
SIP INVITE

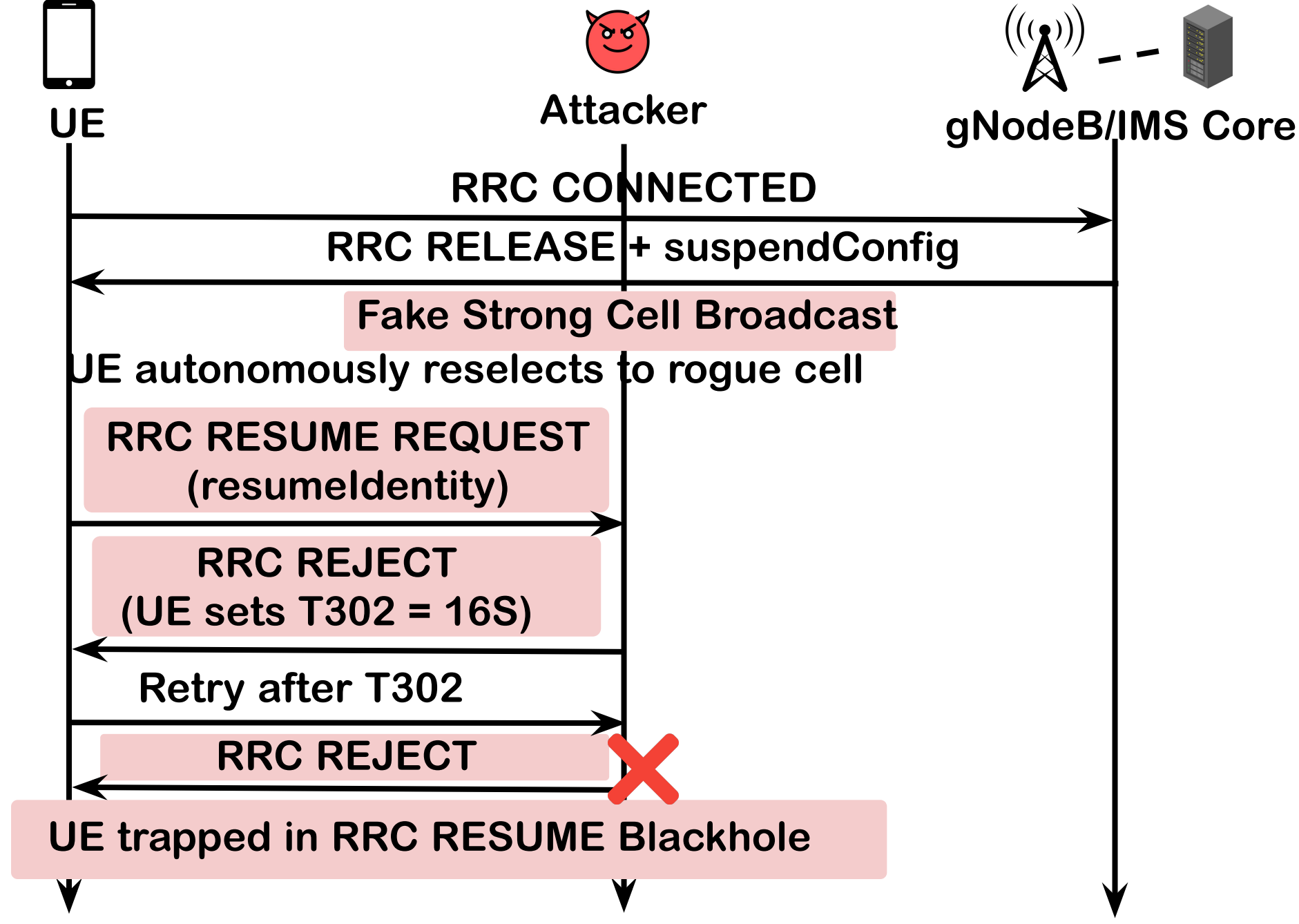

UE
Attacker
gNodeB/IMS Core
Monitor the unencrypted emergency signaling
REGISTRATION REQUEST (emergency)
REGISTRATION ACCEPT
RRC RELEASE
Teardown of the UE radio connection for emergency services
Attack 1
Unencrypted SIP INVITE
SIP BYE
Emergency call ended
Attack 2
EMERGENCY
Attacks 1 & 2 prevent the UE from using emergency services


UE
Attacker / DNS
Non-legitimate endpoint
Legitimate N3IWF
DNS Query (PLMN discovery)
Poisoned DNS response
DNS resolution points to non-legitimate endpoint
Attempt to connect to resolved endpoint
UE cannot reach legitimate N3IWF, Non-3GPP access registration cannot complete


Caller
UE
IMS
Core
Callee
UE
SIP INVITE
Forward signaling to callee
SIP INVITE received
User-Agent header exposed
(phone model + patch ver.)
SIP SESSION PROGRESS
IMS adds signaling headers during interworking
Added headers:
Callee's identity and
IMS server information


UE
Attacker
eNodeB/IMS Core
ATTACH REQUEST
ATTACH ACCEPT
Paging for CS call
EXTENDED SERVICE REQUEST
Sniff unciphered
Forged SERVICE REJECT
(Cause #39: CSFB not
available)
(T3442 = 30 min)
UE accepts reject, CSFB
aborted, UE starts T3442
backoff timer


UE
Attacker
gNodeB/eNodeB
RRC CONNECTED
RRC RELEASE + suspendConfig
Rogue cell advertises stronger signal
UE autonomously reselects to rogue cell
Rogue cell broadcasts
SIB1 (5G) / SIB2 (4G)
ac-BarringFactor = 0
UE decodes SIB barring information
MO access category is barred
UE does not initiate RRC connection
UE cannot access network


UE
Attacker
gNodeB/IMS Core
RRC CONNECTED
RRC RELEASE + suspendConfig
Fake Strong Cell Broadcast
UE autonomously reselects to rogue cell
RRC RESUME REQUEST
(resumeIdentity)
RRC REJECT
(UE sets T302 = 16S)
Retry after T302
RRC REJECT
UE trapped in RRC RESUME Blackhole